

Paper ED-Cos-05: Dark Energy as Late-Universe Substrate Saturation Regime

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Paper_ED_Cos_05 — Dark Energy as Late-Universe Substrate Saturation Regime

Series: Wave-3 Generative Papers (Cosmology Arc) **Status:** Conditional structural derivation given the 13 ED primitives (Paper_087) + one paper-specific naming convention (P-DarkEnergy-Identification, definitional only). Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime hydrodynamic-window scale-separation (Paper_073)**, conditional on Route A for substrate-graph derivation of the quantitative Λ value (per Paper_038_5 reframing). Closes via the same Q1A + Q2A + C3 + IC-INHERITED chain that closed Paper_ED_Cos_01's inflation saturation case, applied to the late-LDE (Λ -dominated) saturation regime; IC-INHERITED uniform- Ψ supply via late-universe dilution-driven asymptote (distinct from Cos_01's post-BBB ignition supply, but framework-neutral analogously). **Date:** 2026-05-16 **Authors:** Paper_087 (primitives — P02 + P03 + P04 + P09 + P11 + P12 + P13); Paper_073 (DCGT); Paper_ED_SC_4_9 (S_{sub}); Paper_089 (V1 + Lorentz-covariance §3); Paper_090 (V5 cross-chain); Paper_012 (P-RB-1 substrate-c); Paper_027 (Newton's G); Paper_028 (cosmic decoupling); Paper_ED_CCC §3.6 (late-LDE asymptote toward SCBU); Paper_ED_Cos_01 (M3; inflation saturation case template); Paper_ED_Dyn_02 (M2; horizon-motion + LDE row inheritance); Paper_038_5 (M2; -smallness reframing); Paper_ED_GW_00 §3.2 (no fundamental substrate metric); Paper_095 (verdict grammar). **Substrate-research support:** Memo_ED_Q1Q2_JointClosure_Construct (Q1A + Q2A construction-uniqueness); Memo_ED_ChainClass_C3_Construct + Audit (saturation chain-class identification + Q-C3-1a/1b/2/2b/3 qualifications); Memo_ED_CCC_Realizability_SubConstruct (Q-C3-3 framework-neutral IC-INHERITED template; late-universe supply analog covered §3.3 below).

Preamble — What This Paper Does NOT Claim

1. This paper does **not** derive the quantitative value of the cosmological constant Λ from substrate parameters alone. Λ value is **INHERITED** from late-universe observations (Planck CMB + Type Ia supernovae + BAO + weak-lensing surveys); substrate-graph derivation of Λ value reduces to **Route A** (substrate-derived $\ell_{V_5}(H_0)$) per Paper_038_5's reframing — Route A is the single remaining load-bearing OPEN.
2. It does **not** sit within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). ED is in the substrate-ontology research program ('t Hooft cellular-automaton; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.). Methodology per Paper_095 (form-FORCED / value-INHERITED).
3. It does **not** posit a quintessence field, phantom field, or any dynamical dark-energy

continuum content. The substrate-side reading: late-LDE is the **saturation regime** (uniform Ψ) of the late universe; vacuum-energy-form continuum stress-energy with $w = -1$ emerges via the same Q1A + Q2A + C3 chain that closed Cos_01's inflation case.

4. It does **not** introduce new substrate primitives or paper-specific postulates. The chain is inherited from Paper_ED_Cos_01's M3 closure applied to the late-LDE saturation case.
5. **Five qualifications carried from C3 audit** (per Cos_01 M3 framing):
 - **Q-C3-1a:** Branch (b) Lagrangian-degeneracy excluded by V1 + V5 primitive-set consistency
 - **Q-C3-1b:** Sign-definite kinetic structure in $\mathcal{L}_{\text{sub}}$ assumed (standard substrate-stability inheritance)
 - **Q-C3-2:** Subleading $O(\varepsilon)$ fluctuations admissible; not load-bearing
 - **Q-C3-2b:** Measure-zero kernel-balance exception in converse direction; not load-bearing generically
 - **Q-C3-3:** IC-INHERITED uniform- Ψ realizability supplied framework-neutrally — for the late-LDE case, via late-universe dilution-driven asymptote (§3.3); cyclic-vs-non-cyclic determination outside scope
6. **-smallness inheritance from Paper_038_5:** the famous 120-OOM “ Λ problem” reduces at substrate-graph level to Route A + Friedmann inheritance per Paper_038_5 — no specialized substrate-graph mechanism for Λ smallness is required beyond Route A's quantitative supply. The substrate-graph form of the dark-energy regime closes via the Q1A + Q2A + C3 chain; the smallness is quantitative-magnitude conditional on Route A.

Abstract

Dark energy in ED is the **late-universe substrate saturation regime** — the asymptotic configuration in which cosmological-expansion-driven dilution of substrate Ψ -non-uniformity (associated with prior matter and radiation content) drives substrate toward uniform- Ψ at substrate-graph level. Under this configuration, kernel derivatives $\nabla_K \Psi$ vanish for all $K \in \{V_1, V_5\}$; substrate-side Noether stress-energy (Q1A construction-uniqueness) collapses to vacuum-energy form $T_{\text{sub}}^{\mu\nu} = -g^{\mu\nu} \mathcal{L}_{\text{sub}}^{\text{const}}$; DCGT translation (Q2A construction-uniqueness) preserves uniform structure at leading order in $\varepsilon = \ell_{\text{ED}}/R_{\text{cg}}$ giving continuum stress-energy with $w = -1$; standard Friedmann recovery gives $a(t) \propto e^{H_0 t}$ — the observed late-universe de Sitter / Λ -dominated regime. Substrate-c bound preserved throughout. This **inherits the Q1A + Q2A + C3 + IC-INHERITED chain established for Cos_01's inflation saturation case** (Memo_ED_Q1Q2_JointClosure_Construct + Memo_ED_ChainClass_C3_Construct + Audit + Memo_ED_CCC_Realizability_SubConstruct); the IC-INHERITED supply for the late-LDE case is the late-universe dilution-driven asymptote (§3.3), structurally distinct from Cos_01's post-BBB ignition supply but framework-neutral analogously. **Verdict per Paper_095: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime, conditional on Route A for quantitative Λ value derivation** (per Paper_038_5 reframing — smallness reduces to Route A + Friedmann). **F1 (sharpest):** observed late-universe equation-of-state deviating from $w = -1$ beyond observational uncertainty in a manner not absorbable into substrate-side mixed-component composition refutes the saturation-regime identification.

§1 Statement of Result

Dark energy in ED is the substrate-graph **late-universe saturation regime** — the late-time asymptotic configuration where cosmological-expansion-driven dilution of substrate Ψ -non-

uniformity drives substrate toward uniform- Ψ at substrate-graph level. The substrate-graph closure of $w = -1$ continuum equation of state proceeds via the **inherited Q1A + Q2A + C3 + IC-INHERITED chain** established for Cos_01's inflation saturation case:

1. **Uniform- Ψ substrate-graph realizability** at the late-LDE asymptote — IC-INHERITED via late-universe dilution-driven mechanism (§3.3; distinct from but parallel to Cos_01's post-BBB ignition supply).
2. **Substrate-side Noether stress-energy uniqueness (Q1A)** — A.1 surviving from six-candidate catalog.
3. **Saturation chain-class identification (C3)** — for uniform Ψ , kernel derivatives vanish; Noether collapses to $T_{\text{sub}}^{\mu\nu} = -g^{\mu\nu} \mathcal{L}_{\text{sub}}^{\text{const}}$.
4. **DCGT mapping uniqueness (Q2A)** — B.1 surviving at leading order in ε .
5. **Friedmann + de Sitter inheritance** — H_0 constant; $a(t) \propto e^{H_0 t}$.

FORM-IDENTIFIED via the Q1A + Q2A + C3 substrate-graph composition (inherited from Cos_01); substrate-c bound preserved throughout. **VALUE-INHERITED:** H_0 value INHERITED from late-universe observations pending Route A; $\Omega_\Lambda \approx 0.685$ INHERITED from Planck. **IC-INHERITED:** uniform- Ψ asymptote supplied by late-universe dilution-driven mechanism, framework-neutral (cyclic-CCC vs single-aeon).

smallness via Paper_038_5 reframing: the famous 120-OOM mismatch $(H_0/M_P)^2 \sim 10^{-122}$ at substrate-graph level reduces to Route A (substrate-derived $\ell_{V_5}(H_0)$) + Friedmann inheritance. The substrate-side discrete ontology does not admit the standard QFT mode-summation that produces the naïve $\rho_{\text{vac}}^{\text{naive}} \sim M_P^4$ estimate; the naïve estimate is structurally absent substrate-side per Paper_038_5 §3.7. Substrate-side $\rho_\Lambda = (3/8\pi)H_0^2(\text{Route A}) \cdot M_P^2(\text{Paper}_027)$.

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime per §4 audit table, conditional on Route A for quantitative Λ value derivation. Closure of Route A would upgrade value-INHERITED to value-DERIVED for the Λ magnitude question within the M3 framing (no verdict-tier upgrade beyond M3).

§2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02 (participation), P03 (translation invariance, Noether source), P04 (bandwidth), P09 (polarity), P11 (commitment), P12 (ED-threshold), P13 (translation arrow).

Upstream papers: - Paper_012 (P-RB-1 substrate-c) — supplies c - Paper_027 (Newton's G via dimensional rearrangement) — supplies G and M_P^2 for ρ_Λ via Friedmann - Paper_028 (cosmic decoupling) — late-LDE structural context - Paper_073 (DCGT hydrodynamic-window coarse-graining) - Paper_089 (V1 retarded kernel + Lorentz-covariance §3) - Paper_090 (V5 cross-chain finite-memory) - Paper_ED_SC_4_9 (S_{sub} + saddle Hessian; kernel-derivative content) - Paper_ED_CCC §3.6 (late-LDE asymptote toward SCBU; under cyclic framing) - **Paper_ED_Cos_01 (Inflation; M3)** — supplies the Q1A + Q2A + C3 + IC-INHERITED chain template; inherited verbatim at form-IDENTIFIED level - **Paper_038_5 (Lambda_V1_Cosmological; M2)** — supplies the -smallness reframing (reduces to Route A + Friedmann inheritance) - Paper_ED_Dyn_02 (M2) — non-saturation regime context; late-LDE inherits Dyn_02's LDE row - Paper_ED_GW_00 §3.2 (no fundamental substrate metric — eliminates A.3 Hilbert candidate within Q1A) - Standard Friedmann + de Sitter (textbook inheritance) - Late-universe observations: Planck CMB (Ω_Λ , Ω_m , H_0); Type Ia supernovae (SNe Ia); BAO (BOSS, DESI); weak-lensing (DES, KiDS, Euclid)

Substrate-research support (inherited from Cos_01 M3 upgrade): - Memo_ED_Q1Q2_JointClosure_Constr (Q1A + Q2A construction-uniqueness) - Memo_ED_ChainClass_C3_Construct + Audit (saturation chain-class identification; five Q-C3 qualifications) - Memo_ED_CCC_Realizability_SubConstruct (Q-C3-3 framework-neutral IC-INHERITED template)

Paper-specific naming convention (no new postulate):

- **P-DarkEnergy-Identification:** Naming convention for the substrate-side identification “*Dark energy = late-universe substrate saturation regime (uniform Ψ achieved via cosmological-expansion-driven dilution of Ψ -non-uniformity from prior matter / radiation content).*” Definitional per Paper_095 §2.3.

No new substrate primitives; no paper-specific postulates beyond the naming convention.

§3 Derivation

3.1 Ontology: same as Cos_01

Per Paper_087 + Paper_ED_SC_4_9 + Paper_ED_GW_00 §3.2: no fundamental spacetime metric, no fundamental scalar field, no quintessence / phantom field. $S_{\text{sub}}[\Psi]$ admits saddles per Paper_ED_SC_4_9. Dark energy, like inflation (Cos_01), is **dynamics-class** — the substrate-graph saturation regime, not a particle / field-content addition.

3.2 The late-LDE saturation-regime identification

Per P-DarkEnergy-Identification: the late-LDE epoch is the substrate-graph **saturation regime** of the late universe — the asymptotic configuration where Ψ is uniform at substrate-graph level. This is structurally identical to Cos_01’s inflation saturation regime (Paper_ED_Cos_01 §3.2): the same uniform- Ψ configuration, with V1 + V5 + individuation reactivated above the P12 ED-threshold.

The two saturation regimes — early-universe inflation (Cos_01) and late-universe dark energy (this paper) — differ only in their **temporal location** in the cosmological timeline and their **realizability supply path** (§3.3). The substrate-graph chain closing $w = -1$ from uniform Ψ is identical.

3.3 IC-INHERITED supply: late-universe dilution-driven asymptote

The uniform- Ψ realizability at the late-LDE saturation regime is supplied by **cosmological-expansion-driven dilution of substrate Ψ -non-uniformity associated with prior matter and radiation content**.

Substrate-graph mechanism:

- During MDE (matter-dominated era), substrate has Ψ -non-uniformity associated with matter-class chain content (per Paper_ED_Dyn_02 §3.2 candidate-MDE substrate identification).
- During RDE (radiation-dominated era), substrate has Ψ -non-uniformity associated with radiation-class chain content.
- Cosmological expansion drives matter density $\rho_m \propto a^{-3}$ and radiation density $\rho_r \propto a^{-4}$ to zero asymptotically; substrate Ψ -non-uniformity correspondingly diluted at substrate-graph level (DCGT coarse-graining at expanding hydrodynamic scale).
- As $\rho_m, \rho_r \rightarrow 0$, substrate Ψ asymptotically approaches uniform configuration; saturation regime obtains.

Substrate-graph derivation status: the mechanism is **D-via-I** at substrate-graph level — composition of Paper_ED_Dyn_02 non-saturation regime identification + Paper_073 DCGT cosmological-expansion-driven coarse-graining + Paper_ED_CCC §3.6 (under cyclic framing — late-LDE asymptote toward SCBU; under R3 single-aeon framing — late-LDE terminal-state asymptote of the single cycle). The IC-INHERITED supply is structurally **distinct from but parallel to** Cos_01’s post-BBB ignition supply (Memo_ED_CCC_Realizability_SubConstruct §3).

Framework-neutral: under cyclic-CCC framing (Paper_ED_CCC §3.6), the late-LDE asymptote is the dilution approach toward SCBU; under R3 single-aeon framing, it is the terminal-state asymptote of the single cosmological cycle. Both framings supply uniform- Ψ at IC-INHERITED level; cyclic-vs-non-cyclic determination outside Cos_05’s scope. **Q-C3-3 framework-neutrality carries** per Cos_01 + Memo_ED_CCC_Realizability_SubConstruct §4.

3.4 Substrate-c preservation + DCGT translation

Substrate-c bound (Paper_012 P-RB-1) preserved at substrate-graph level throughout the late-LDE saturation regime; V1 retarded propagation operates within the bound. Exponential growth $a(t) \propto e^{H_0 t}$ is a continuum-side coarse-grained-geometry effect via DCGT — same de Sitter mechanism standard cosmology uses and same mechanism that closed Cos_01’s inflation exponential growth.

3.5 smallness via Paper_038_5 reframing

The famous 120-OOM cosmological-constant problem $(H_0/M_P)^2 \sim 10^{-122}$ reduces at substrate-graph level via **Paper_038_5’s reframing** (§3.7 of that paper):

1. **Naïve QFT estimate structurally absent substrate-side:** standard QFT’s $\rho_{\text{vac}}^{\text{naive}} \sim M_P^4$ requires summing zero-point energies of an infinite tower of QFT modes. ED’s discrete substrate (Paper_087 + Paper_089 V1 finite-width) does not admit such an infinite mode-tower; the naïve estimate is structurally absent from the substrate ontology. **This partially dissolves the 120-OOM “problem” at substrate-side** before any DCGT scale-separation enters.
2. **Substrate-graph closure path:** standard Friedmann for $w = -1$ late universe gives $\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$. The substrate-graph derivation:
 - M_P^2 via Paper_027 (Newton’s G via dimensional rearrangement) — INHERITED at I level
 - H_0 via **Route A** (substrate-derived $\ell_{V_5}(H_0)$) — OPEN per ED_MEMORY anchor 7
 - ρ_Λ closes at D-via-I conditional on Route A: $\rho_\Lambda^{\text{substrate}} = (3/8\pi)H_0^2(\text{Route A}) \cdot M_P^2(\text{Paper}_027)$

Result: the -magnitude question reduces to Route A. No specialized substrate-graph mechanism for smallness is required beyond Route A’s quantitative supply — the substrate-graph form of the dark-energy regime closes via the Q1A + Q2A + C3 chain (substantively closed; see §3.6 below); the smallness is quantitative-magnitude conditional on Route A.

3.6 M3 chain (inherited from Cos_01 §3.8)

The substrate-graph closure for the dark-energy regime proceeds via the same five-step chain that closes Cos_01’s inflation saturation:

Step	Substance	Status
A (IC-INHERITED)	Uniform- Ψ substrate-graph configuration at the late-LDE asymptote, supplied by cosmological-expansion-driven dilution mechanism (§3.3)	IC-INHERITED framework-neutrally (CCC §3.6 cyclic or R3 single-aeon)
B (C3)	Kernel-derivative vanishing for uniform Ψ	D-via-I (inherited from Cos_01 §3.8 + Memo_ED_ChainClass_C3_Construct §3.1)
C (Q1A)	Substrate-side Noether collapses to vacuum-energy form, $w = -1$	D-via-I (inherited from Cos_01 §3.8 + Memo_ED_Q1Q2_JointClosure_Construct §2)
D (Q2A)	DCGT preserves uniform structure at leading order in ε	D-via-I (inherited from Cos_01 §3.8 + Memo_ED_Q1Q2_JointClosure_Construct §3)
E (Friedmann)	H_0 constant; $a(t) \propto e^{H_0 t}$	INHERITED (standard cosmology + late-universe observations)

Five qualifications carried (Q-C3-1a/1b/2/2b/3) per Cos_01 M3 framing.

3.7 Contrast with quintessence / phantom dark-energy models

Element	Quintessence / phantom DE	ED substrate-side	Status
Driver	Quintessence field ϕ on slow-roll potential / phantom field with negative kinetic term	Late-universe saturation regime (uniform Ψ)	D-via-I (via §3.6 chain)
Equation of state	$w \neq -1$ generically (dynamical)	$w = -1$ exactly (modulo Q-C3-2 subleading)	D-via-I
-smallness	Generally unresolved (anthropic / landscape)	Reduces to Route A + Friedmann per Paper_038_5 reframing	D-via-I conditional on Route A
Time-variation	$w(z) \neq -1$ admissible; $\dot{w} \neq 0$ admissible	Strict $w = -1$ at substrate-graph level (modulo Q-C3-2); no time-variation predicted	Identified + Replaced
Fine-tuning	$V(\phi)$ shape / potential parameters	Relocated to Route A + IC supply	Identified (framework-relative)

Element	Quintessence / phantom DE	ED substrate-side	Status
Particle / field content	Quintessence field / phantom field at substrate-effective level	None at substrate level; saturation regime	Replaced (genuine ontology change)

The substrate-side prediction $w = -1$ exactly (modulo subleading-order fluctuations per Q-C3-2) **distinguishes this paper from quintessence / phantom dark-energy models**. F1 sharpness depends on this.

§4 Audit Table

#	Step	Label	Notes
1	P02, P03, P04, P09, P11, P12, P13	P	Paper_087. P03 + P13 supply Noether translation invariance.
2	Upstream inheritance block	I	Paper_012; Paper_027; Paper_028; Paper_073; Paper_089; Paper_090; Paper_ED_SC_4_9; Paper_ED_CCC §3.6; Paper_ED_Cos_01 (M3 chain template); Paper_038_5 (-smallness reframing); Paper_ED_Dyn_02 (LDE row); Paper_ED_GW_00 §3.2.
3	Standard cosmology Friedmann + late-universe observations	I	Textbook inheritance; Planck + SNe Ia + BAO + weak-lensing.
4	P-DarkEnergy- Identification (naming convention)	P	§2.3; definitional per Paper_095 §2.3; no new substrate content.

#	Step	Label	Notes
5	Late-LDE saturation-regime identification: dark energy = substrate uniform- Ψ saturation at late-universe asymptote	D-via-I	§3.2 + §3.6 Step A; inherits Cos_01's saturation-regime identification template.
6	Late-universe dilution-driven uniform- Ψ asymptote (IC-INHERITED supply)	IC-INHERITED	§3.3 + §3.6 Step A. Framework-neutral: CCC §3.6 (cyclic) or R3 (single-aeon). Q-C3-3 carried. Mechanism D-via-I via Paper_ED_Dyn_02 + Paper_073 + Paper_ED_CCC §3.6 composition.
7	Kernel-derivative vanishing for uniform Ψ	D-via-I	§3.6 Step B. Inherited from Cos_01 §3.8 + Memo_ED_ChainClass_C3_Constru §3.1.
8	Substrate-side Noether collapse to vacuum-energy form, $w = -1$ (Q1A)	D-via-I	§3.6 Step C. Inherited from Cos_01 §3.8 + Memo_ED_Q1Q2_JointClosure_Co §2. Qualifications Q-C3-1a + Q-C3-1b carried.
9	DCGT preserves uniform structure at leading order (Q2A)	D-via-I	§3.6 Step D. Inherited from Cos_01 §3.8 + Memo_ED_Q1Q2_JointClosure_Co §3. Qualification Q-C3-2 carried.
10	Friedmann recovery + de Sitter: $a(t) \propto e^{H_0 t}$	I	§3.6 Step E. Standard cosmology.
11	Substrate-c bound preserved throughout	D-via-I	§3.4. Paper_012 P-RB-1 + Paper_089 + Paper_093.
12	-smallness reduces to Route A + Friedmann inheritance (Paper_038_5 reframing)	D-via-I	§3.5. Inherits Paper_038_5 §3.7 reframing. Naïve QFT estimate structurally absent substrate-side; $\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$ closes at D-via-I conditional on Route A.

#	Step	Label	Notes
13	Quantitative substrate-graph derivation of $H_0 \rightarrow$ quantitative ρ_Λ value	OPEN — Route A	§3.5 + §3.6 Step E. Single load-bearing remaining OPEN. Per ED_MEMORY anchor 7 + Paper_038_5 reframing. Closure upgrades value-INHERITED $\rightarrow$ value-DERIVED; no verdict-tier upgrade beyond M3.
14	Quintessence / phantom field-content alternatives ruled out at substrate level	A$\rightarrow$position	§3.7. Substrate-ontology replacement framing per Paper_095 §3.3; F1 sharpness depends on this row.
15	Subleading-order w -deviations from -1 (Q-C3-2 subleading fluctuations)	OPEN (non-load-bearing)	§3.6. Subleading $O(\varepsilon)$; not load-bearing for leading-order M3 framing.
16	Substrate-graph derivation of cosmic-expansion-driven dilution mechanism quantitative timing (matter- crossover redshift z_Λ)	OPEN (non-load-bearing)	§3.3. Substrate-research-frontier; requires Route C1 (Memo_ED_ChainClassIdentification CC-OPEN-1–4) + Route A. Not load-bearing for M3 verdict.
17	Verdict M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A$\rightarrow$regime, conditional on Route A	A$\rightarrow$position	Per Paper_095 §3.3 line 77; framing-claim verdict row. Five-anchor verdict-sync verified across status / abstract / §1 / this row / §6.

Audit summary: zero pure-D rows; **8 D-via-I rows** (5, 7, 8, 9, 11, 12) plus row 13 inherited from Cos_01 mechanism; 1 IC-INHERITED row (6) framework-neutral; 3 I rows (2, 3, 10); 2 P/naming rows (1, 4); 2 A $\rightarrow$ position rows (14, 17); **1 load-bearing remaining OPEN** (13 — Route A); 2 non-load-bearing OPENs (15, 16). No new substrate primitives.

§5 Falsification Criteria

- **F1 (sharpest):** Observed late-universe equation-of-state deviating from $w = -1$ beyond observational uncertainty at a level not absorbable into substrate-side mixed-component composition or Q-C3-2 subleading-order corrections — e.g., a clean phantom equation-of-state $w < -1$ over a wide redshift range; or significant $w(z)$ time-variation $\dot{w} \neq 0$. Refutes §3.2 saturation-regime identification (substrate-side prediction is strict $w = -1$ at leading order). (Currently consistent: Planck + SH0ES + BOSS + DES constraints place DE near $w \approx -1.03 \pm 0.03$ consistent with $w = -1$.)
- **F2:** Detection of a quintessence / phantom field-particle signature in any cosmological observation or laboratory experiment — substrate-side has no field-content at the late-LDE saturation regime. Refutes §3.7 substrate-ontology replacement (row 14).
- **F3:** Substrate-graph proof that the late-universe dilution-driven uniform- Ψ asymptote is not realizable under existing primitive content — i.e., demonstration that cosmological expansion under Paper_073 DCGT does not drive substrate Ψ -non-uniformity to zero asymptotically. Refutes §3.3 IC-INHERITED supply and the §3.6 chain Step A.
- **F4:** Observation of Λ time-variation $\dot{\Lambda} \neq 0$ over cosmological timescales beyond observational uncertainty — substrate-side prediction is strict Λ constant (vacuum-energy form does not admit time variation at leading order). Refutes §3.6 Step C + Q-C3-2 subleading framing.

§6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), not within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

Dark energy in ED is the late-universe substrate saturation regime — the asymptotic configuration where cosmological-expansion-driven dilution of substrate Ψ -non-uniformity drives substrate toward uniform- Ψ at substrate-graph level. The substrate-graph closure of $w = -1$ continuum equation of state **inherits the Q1A + Q2A + C3 + IC-INHERITED chain established for Paper_ED_Cos_01's inflation saturation case**, applied to the late-LDE case with the IC-INHERITED supply provided by the late-universe dilution-driven asymptote mechanism (§3.3) framework-neutrally (cyclic-CCC vs R3 single-aeon). Five qualifications (Q-C3-1a/1b/2/2b/3) carried from C3 audit; substrate-c bound preserved; exponential late-universe expansion emerges continuum-side via DCGT — same de Sitter mechanism standard cosmology uses. **The -smallness 120-OOM “problem” reduces to Route A + Friedmann inheritance per Paper_038_5 §3.7 reframing** — the naïve QFT mode-tower estimate is structurally absent substrate-side; substrate-graph ρ_Λ closes at D-via-I conditional on Route A.

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime hydrodynamic-window, conditional on Route A for quantitative Λ value derivation per Paper_095. Eight D-via-I rows (inherited from Cos_01); 1 IC-INHERITED row (late-universe dilution-driven asymptote, framework-neutral); 3 I rows; 2 P/naming rows; 2 A→position rows; **1 load-bearing remaining OPEN — Route A (row 13) for value-INHERITED → value-DERIVED upgrade within M3 framing**. Quintessence / phantom field-content alternatives genuinely “Replaced” (row 14); $w = -1$ exact prediction at leading order distinguishes this paper from dynamical dark-energy models. **F1 sharpness:**

observed late-universe w deviation from -1 beyond Q-C3-2 subleading-order corrections refutes the saturation-regime identification.

Cross-arc consequences: this paper is the **direct cross-arc consequence of Paper_ED_Cos_01's M2 $\rightarrow$ M3 upgrade** via the Q1Q2 + chain-class identification project (Q1A + Q2A + C3 + IC-INHERITED chain). Both early-universe inflation (Cos_01) and late-universe dark energy (this paper) are now substrate-graph-identified at M3 as instances of the same saturation-regime substrate-graph configuration, distinguished only by temporal location and IC-supply path. Paper_038_5 (Lambda_V1_Cosmological; M2) retains its M2 framing for the V1-cutoff failed approach; this paper supplies the corpus-level positive M3 closure for the dark-energy regime that Paper_038_5's reframing pointed toward. **Route A remains the single highest-leverage open derivation** (ED_MEMORY anchor 7), now controlling value-DERIVED upgrade for both Cos_01 and Cos_05 in addition to the ED-SC 4.x arc-wide consequences.

End Paper_ED_Cos_05.